

Understanding Hypertension Risk: IS 204 Final Paper

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Abstract

Hypertension is a widespread global health issue and a leading cause of early mortality, making it a crucial area of study. Our research draws upon the work of Nerenberg et al. (2018) who provide a comprehensive overview of hypertension diagnosis guidelines, which include metrics for risk assessment and management strategies. One specific question we investigated was whether cholesterol levels and BMI were associated with hypertension classification. To explore this, we conducted t-tests to compare cholesterol and BMI between high-risk and low-risk groups and used logistic regression models to classify individuals as at-risk. We analyzed the statistical significance of each factor's contribution to hypertension classification. Our results found no statistically significant associations between the examined risk factors and hypertension risk within our dataset. These findings highlight the complexity of predicting hypertension and underscore the need for more comprehensive datasets and multifactorial models to improve real-world predictive accuracy. Since hypertension often develops without noticeable symptoms, early detection and prevention strategies remain essential. A deeper understanding of contributing risk factors may help reduce the incidence of hypertension and improve public health outcomes and life expectancy.

Introduction

Hypertension, more commonly known as high blood pressure, is a significant global health concern and a leading risk factor for severe medical conditions such as cardiovascular disease, stroke, and kidney failure. An estimated 1.28 billion individuals aged 30–79 worldwide suffer from hypertension, with approximately two-thirds residing in low- and middle-income

countries (World Health Organization, 2023). More alarmingly, 46% of these affected individuals are unaware of their condition. Hypertension is diagnosed when blood pressure readings consistently exceed 140/90 mmHg, making it a primary contributor to premature mortality. Our research focuses on hypertension and identifying the health factors influencing its development. Understanding the risk factors associated with hypertension can aid in developing predictive machine-learning models that assess population-wide risk levels and offer individualized predictions.

Literature Review

To effectively predict hypertension risk, it is crucial to identify key factors and examine methodologies from previous research on hypertension. Nerenberg et al. (2018) provide a comprehensive overview of Canada's hypertension diagnosis guidelines, incorporating blood pressure thresholds, cardiovascular risk assessments, and management strategies. Such management strategies include dietary changes, physical activity, and pharmacological interventions. Similarly, Ibrahim et al. (2012) discussed disparities in hypertension prevalence and management across developing countries, emphasizing the stark impact of socioeconomic and environmental factors. Their study categorizes hypertension risk factors into non-modifiable (e.g., genetic predisposition, ethnicity) and modifiable (e.g., diet, obesity, smoking), underscoring the need for public health interventions to improve treatment accessibility.

In the domain of machine learning applications for hypertension prediction, Silva et al. (2022) explore the integration of artificial intelligence in hypertension research, focusing on dataset processing, programming languages (R and Python in particular), and methodologies such as feature selection, train-test splitting, and data balancing. They highlight the predominance of binary classification in hypertension prediction models, though note that some

research adopts a multi-class approach that includes normotension, prehypertension, and different hypertension stages. Amaratunga et al. (2020) have investigated AI applications in hypertension prediction, evaluating algorithms such as deep learning, support vector machines (SVMs), and neural networks. Despite advancements in AI, their study reports modest accuracy levels (approximately 56–57%), suggesting that machine learning models require further refinement for reliable clinical implementation. Collectively, the current scope of research underscores the importance of exploring diversity in predictive factors and refining modeling techniques.

By synthesizing insights from prior studies on hypertension risk factors and machine learning applications, our study aims to address five key questions regarding hypertension prediction and classification. Firstly, we want to identify which country has the highest proportion of high-risk individuals, hypothesizing that low- and middle-income countries will have a higher prevalence due to limited healthcare access. Second, we intend to investigate whether smoking significantly increases the likelihood of hypertension, predicting a strong positive correlation due to the strain smoking places on the body. We also hope to divide our dataset into distinct age ranges to pinpoint the most vulnerable populations for hypertension diagnosis, expecting that individuals over the age of fifty will show the highest risk classification due to age-related vascular changes. Moving on to lifestyle factors, we then ask whether higher stress levels and reduced sleep duration significantly contribute to hypertension risk, hypothesizing that stress and inadequate sleep will be associated with a higher risk of hypertension. Lastly, we plan to assess the potential associations between gender, cholesterol levels, BMI, diabetes, and hypertension classification, predicting that individuals with higher cholesterol and BMI will be more likely to be classified as high risk for hypertension. By

leveraging machine learning techniques, we aim to ensure improved accuracy in identifying individuals who may be at risk, as well as contribute to more effective prevention strategies.

Methods

Our study sought to investigate five research questions related to hypertension classification using statistical modeling. We chose logistic regression as the primary modeling technique due to its ability to perform binary classification tasks. The dependent variable in all models was hypertension classification (high risk or low risk). The independent variables varied by question but included continuous measures (e.g., age, BMI, cholesterol, stress level, sleep duration) and categorical groupings (e.g., smoking status, country). Transformations included binary encoding for categorical variables and binning continuous variables (e.g., age groups) where appropriate.

Since logistic regression was used throughout, all models shared several underlying assumptions, which we evaluated before analysis. These included linearity in the logit for continuous predictors, absence of multicollinearity (checked using correlation matrices and Variance Inflation Factor), and sufficient sample size relative to predictors. All hypothesis tests conducted in the study were two-tailed, and results were evaluated at an alpha level of 0.05.

To answer our first research question—identifying which countries have the highest proportion of high-risk individuals—we grouped the data by country and computed the average binary hypertension risk score. The binary transformation encoded the original hypertension variable as 1 for high risk and 0 for low risk, producing an estimate of the proportion of high-risk individuals by country. While this question did not involve predictive modeling, it provided important population-level context for the remainder of the study.

Our second research question examined whether age is associated with increased hypertension risk. Age was used both as a continuous predictor and binned into age groups (18–29, 30–39, 40–49, 50–59, 60–69, 70–79, 80–89) to better visualize potential patterns. A logistic regression was fit using continuous age as the independent variable. Before modeling, we used bar charts to explore how hypertension risk varied across age categories.

The third research question asked whether smoking behavior significantly increases the likelihood of hypertension, when controlling for age group. We used logistic regression, with smoking status numerically encoded and age group treated as ordinal categories. Descriptive statistics and visualizations helped us explore the relationship between smoking, age, and hypertension risk.

Our fourth question focused on whether high stress and low sleep duration contribute to hypertension classification. Both variables were first explored through basic EDA. Stress (ordinal, scale 1–9) and sleep (continuous, in hours) were treated as continuous predictors in a multivariate logistic regression model. We briefly tested for interactions between predictors, but these terms did not greatly improve model fit and were thus not included in the final model.

The final research question examined the relationship between cholesterol level, BMI, and hypertension classification. A logistic regression model was built using cholesterol and BMI as continuous predictors. Model performance was evaluated using classification accuracy. We also conducted two-tailed t-tests ($\alpha = 0.05$) to test whether cholesterol and BMI differed significantly between high-risk and low-risk groups. All data cleaning, transformation, and analysis were conducted in Python (version 3.11) using the pandas library for data management, numpy for numerical operations, matplotlib for visualizations, and scikit-learn for logistic modeling.

Discussion & Results

Our sample dataset contained an even split of males and females, with 49.89% male and 50.11% female participants as well as logged information on whether patients have diabetes. The diabetes split looked similarly even, with 50.14% of patients not having diabetes, and 49.86% of patients having diabetes. No other variables indicating prior comorbidities were present in the dataset. The 25th percentile for age was 36 years old, the average age was 54 years old, and the 75th percentile was 71 years old, which all are within the expected range. We utilized these risk factors in our analysis of patient hypertension risk.

To identify which countries showed the highest hypertension risk, we grouped the data by country and counted the number of cases labeled as high-risk for hypertension. The resulting bar chart (Figure 1) revealed that hypertension prevalence was relatively consistent across countries in the dataset, ranging from approximately 6,170 to 6,420 cases per country. Countries such as Saudi Arabia (6,414 cases), South Africa (6,368 cases), and Russia (6,367 cases) appear at the top with the highest number of high-risk hypertension cases. In contrast, countries such as China, Canada, and Argentina recorded the fewest high-risk hypertension cases in the dataset. These results are surprising, as we initially hypothesized that low- and middle-income countries would exhibit the highest hypertension rates due to limited access to preventative healthcare and a lower quality of life. However, these findings do not support this claim. These elevated counts could be attributed to a combination of factors, including lifestyle, dietary habits, genetic predisposition, and a sedentary lifestyle, although further investigation is needed to determine specific causality.

To assess the sole impact of age on hypertension rates, the dataset was divided into seven age groups (18–29, 30–39, ..., 80–89). A bar chart was used to explore age-based differences in

hypertension prevalence (as seen in Figure 2). However, logistic regression using continuous age as a predictor yielded an age coefficient of -0.0005, an odds ratio of 0.9995, and a p-value of 0.398. No multicollinearity was found between age and the dataset's other numerical variables. This result indicates that age does not have a statistically significant effect on hypertension risk. We accept the null hypothesis. The p-value far exceeded the 0.05 threshold, suggesting the observed association may be due to random variation rather than any sort of meaningful pattern.

Our analysis on whether smoking and age group had an increased chance of getting hypertension failed to reject the null hypothesis and conclude that smoking status does not increase the likelihood of hypertension based on age group in this dataset. The logistic regression results indicated that neither smoking status nor age group were significant predictors of hypertension (as seen in Figure 3). The p-value for both smoking status ($p = 0.313$) and age group ($p = 0.151$) showcase that the overall fit was weak. Additionally, the pseudo R-squared ($1.481e-05$) and the likelihood ratio test (LLR p-value: 0.2145) were both low and non-significant.

Using logistic regression to analyze the effect of an individual's stress level on a scale of 1-9 (1 being low and 9 being high) and their average sleep per night, we found that stress level and sleep duration yielded p-values of 0.844 and 0.948, respectively. Given a threshold of 0.05, both p-values proved to be statistically insignificant, indicating sleep and stress do not meaningfully predict hypertension risk in this dataset. Boxplots were used to show the lack of difference in sleep duration or stress levels between individuals classified as high or low hypertension risk (Figures 4 and 5). A heatmap was used to visualize the predicted probability of high-risk hypertension across all combinations of stress and sleep (Figure 6). The predicted

probabilities were around 72% for all categories, further confirming these two predictors have no meaningful effect on hypertension risk in this dataset.

Looking into the relationship between hypertension, cholesterol, and BMI, we similarly found insignificant results (see Figure 7). The model performed with an accuracy level of 0.7174. We also performed t-tests to determine the statistical significance of cholesterol and BMI's impact on hypertension. We used an alpha level of 0.05 as the threshold for significance across all tests. Cholesterol and BMI both did not have a statistically significant impact on hypertension, with p-values of 0.0919 and 0.6712, respectively.

One of the key limitations we encountered in the dataset during our analysis was the lack of clear definitions for the variables (i.e., no codebook was provided). For instance, the stress level indicator did not include explanations for what each numerical value represented. We had to assume that higher values corresponded to higher levels of stress. Additionally, we lacked information on how most measurements were obtained—whether they were self-reported or measured objectively. For example, in the case of sleep data, it was unclear whether participants used a tracking device or simply estimated their average sleep duration. Another issue we discovered during our analysis was an imbalance in the classification of hypertension risk levels: approximately 70% of the dataset was labeled as high-risk, while only 30% was labeled as low-risk. This class imbalance may have biased our models' classification thresholds and reduced generalizability.

For future work, we propose developing a more comprehensive regression model to better predict hypertension risk. Rather than analyzing each risk factor individually, our model would integrate multiple variables to evaluate their combined predictive power. We would also test various machine learning algorithms to identify the most accurate and reliable predictors. A

robust and inclusive model could significantly improve the early identification of individuals at risk, enabling more proactive screening and intervention. Additionally, exploring the relationship between hypertension and comorbid chronic conditions—such as diabetes, heart disease, or kidney disease—could further enhance the model’s practical applications. These relationships may reveal patterns that help identify patients who require more intensive monitoring or tailored treatment plans, making such a model a valuable tool for implementation within real-world healthcare systems.

Conclusion

Our research into the relationship between hypertension and a number of health factors aimed to identify the strongest predictors of hypertension. Our five research questions used statistical modeling to assess the relationship between risk factors (country, age, smoking, stress, etc.) and hypertension risk. Our analysis found no statistically significant associations between the examined risk factors and hypertension risk. This essentially means that each of the relationships we explored showed that the chosen risk factors had no significant impact on hypertension risk. Although our research was inconclusive, understanding risk factors associated with hypertension risk can help create predictive models that assess population-wide risk factors and offer individualized predictions.

References

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Tables & Figures

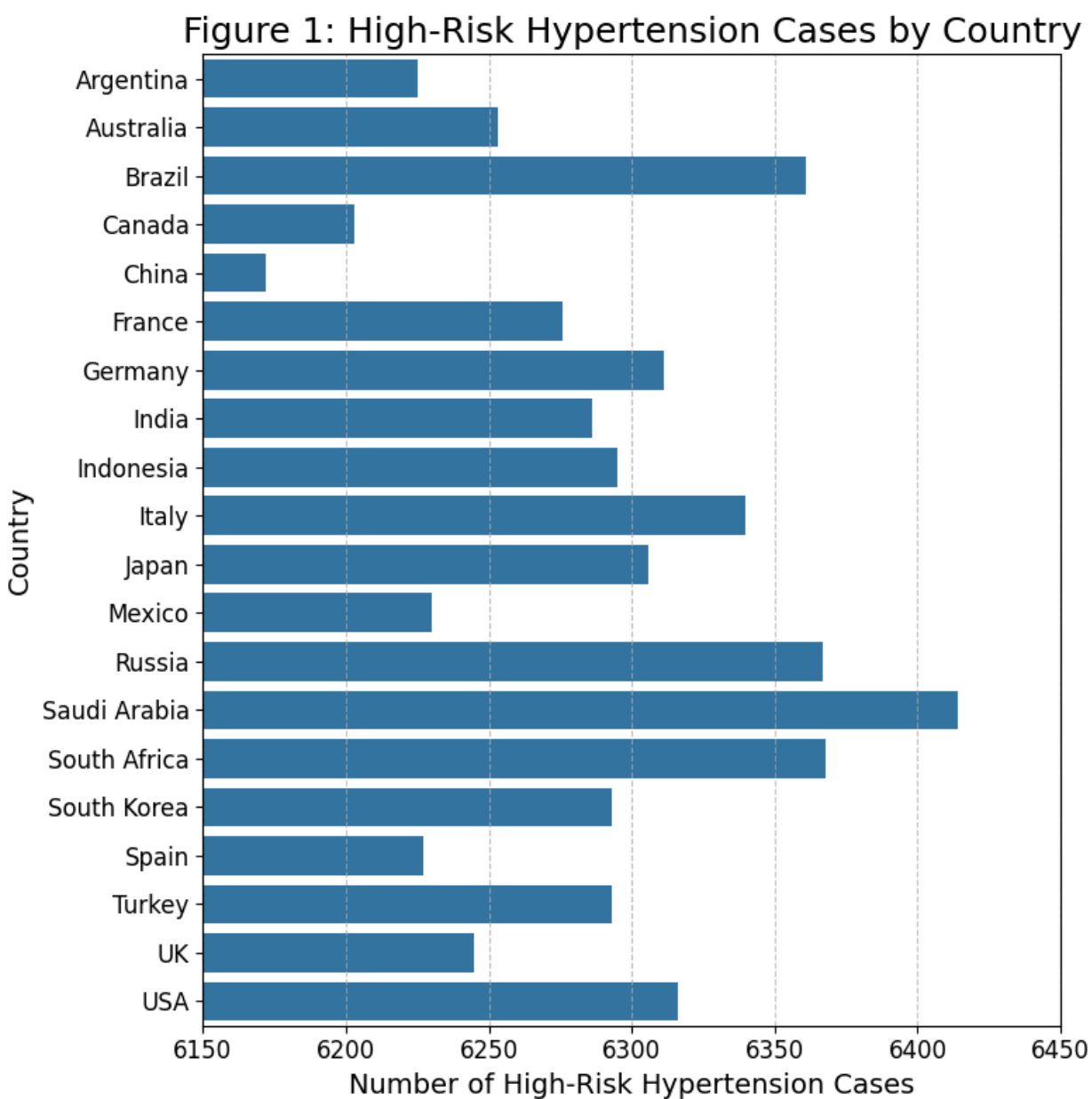


Figure 2: Proportion of Hypertension Risk by Age Group

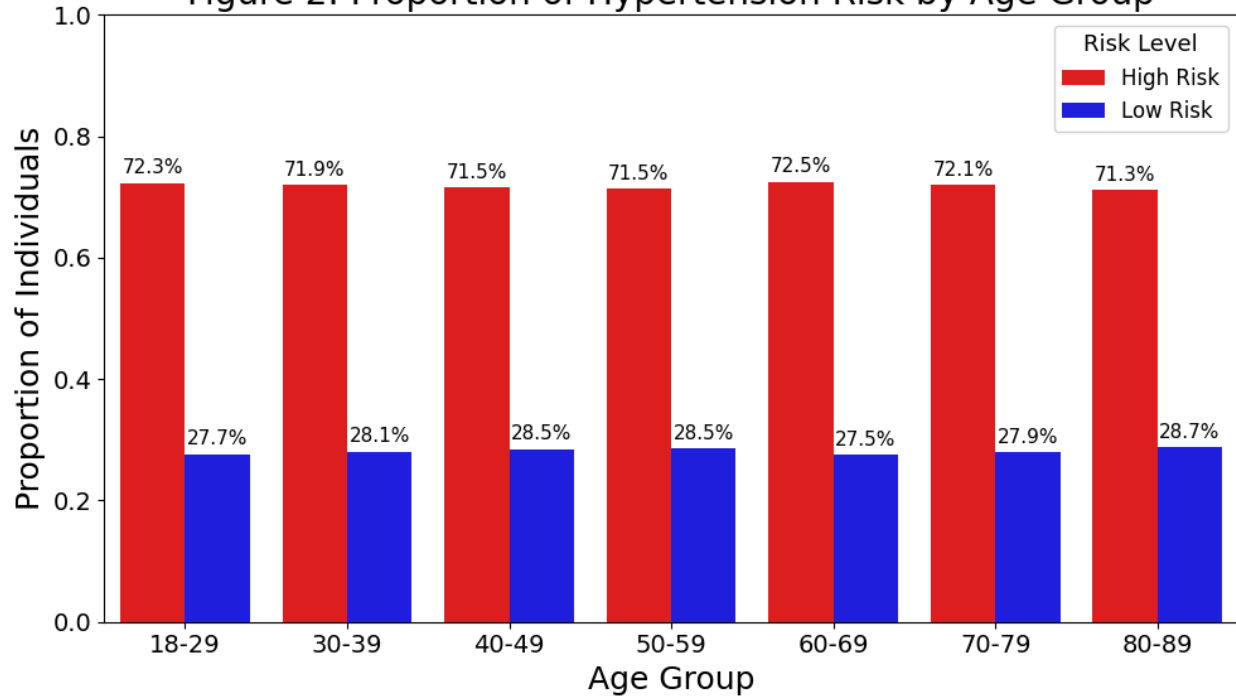


Figure 3: Proportion of High-Risk Hypertension by Age Group and Smoking Status

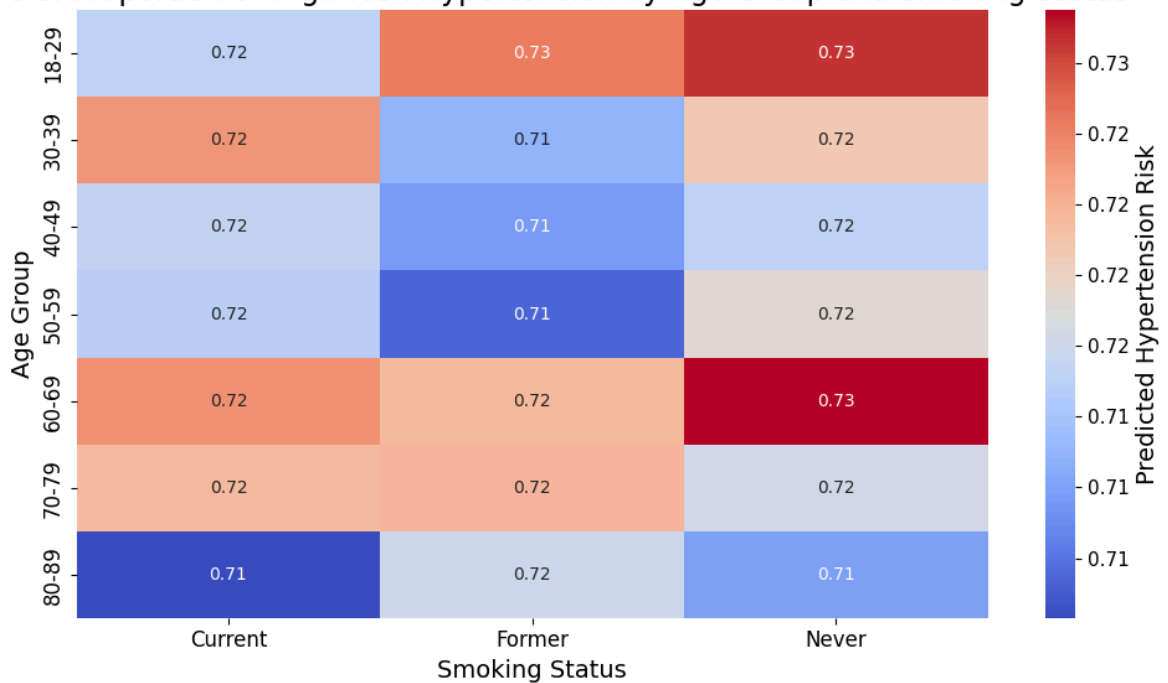


Figure 4: Average Sleep by Hypertension Risk

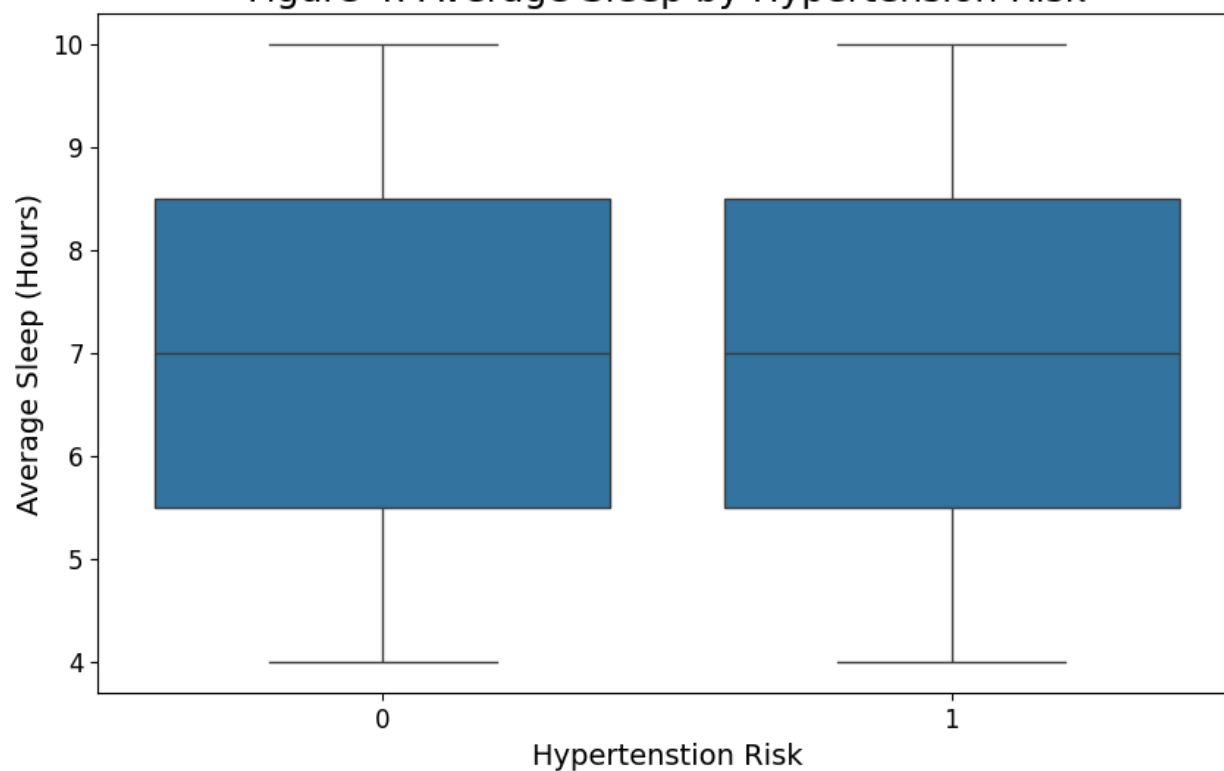


Figure 5: Stress Level by Hypertension Risk

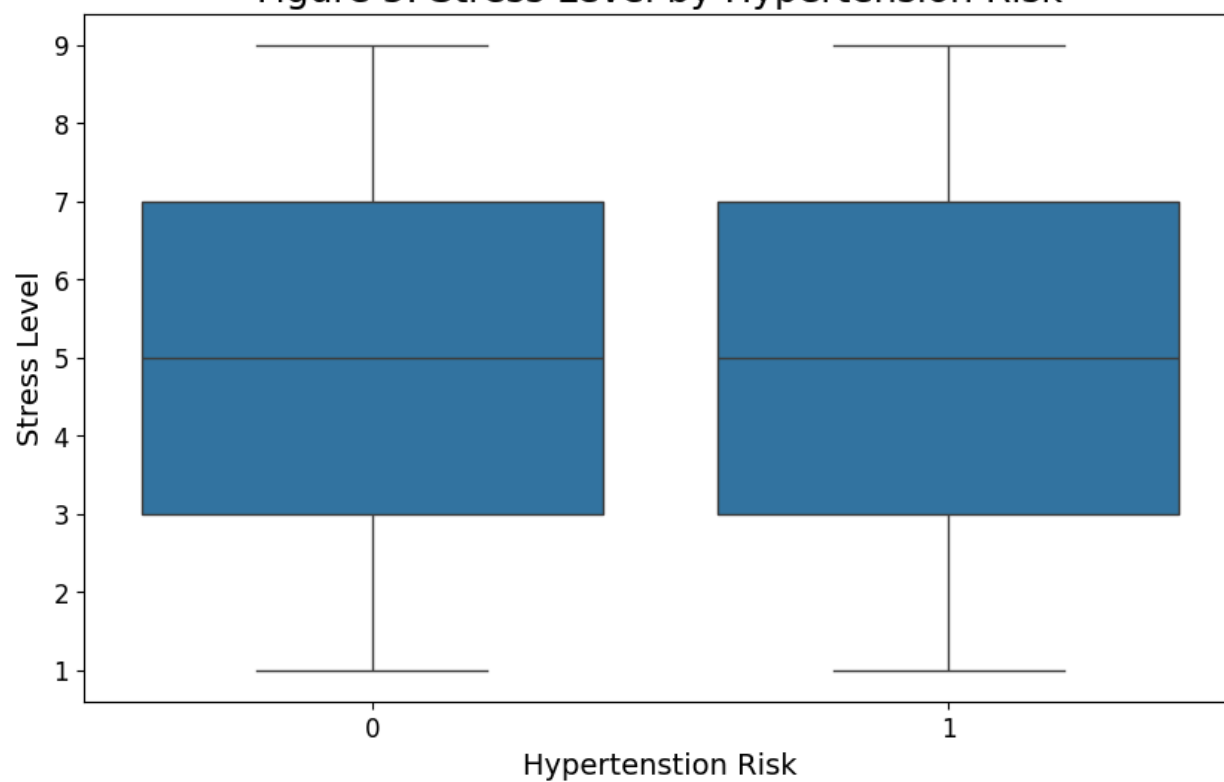


Figure 6: Predicted Hypertension Risk by Sleep and Stress Intervals

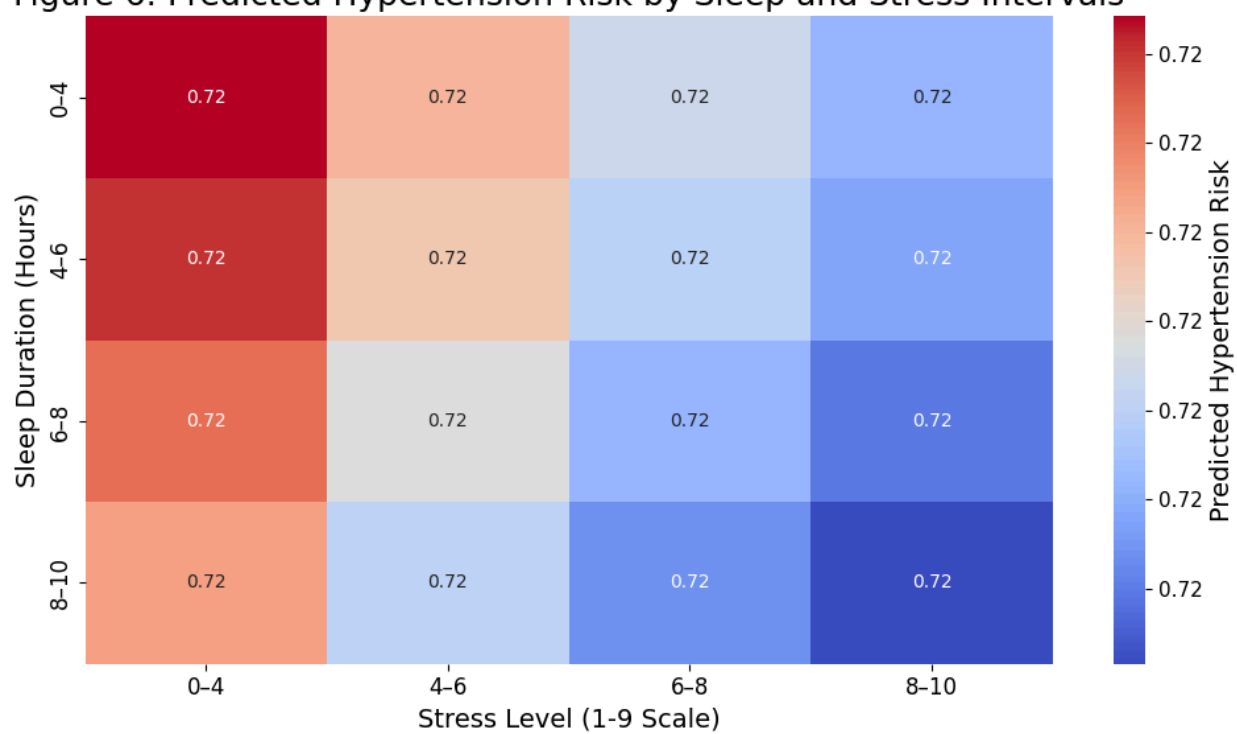


Figure 7: Proportion of Hypertension by Cholesterol and BMI Groups

